3Laws

Release beta

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Nov 02, 2023

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Chapter 1

Introduction

Here is the public documentation for 3LawsRobotics Products:

- Robot Diagnostic Module
- The Supervisor/Copilot

Chapter 2

Robot Diagnostic Module

The **Robot Diagnostic Module (RDM) of 3LawsRobotics** is a powerful tool to gather analytics about your robot. It provides low level metrics such as CPU usage, signal delays but also high level safety critical metrics like safety envelope violation, planning inconsistency or sensor damage detection.

All of these metrics can be viewed on your web browser at 3laws.app.

In order to get your account setup and use freely this RDM, please contact contact@3lawsrobotics.com

This product is still at a Beta development stage. New features are developed every month and some breaking changes can still happened. The 3LawsRobotics Team is available to answer your question.

For now only a ROS1, ROS2 interface is available but we intend to extend the offer of possible interfaces in the near future with support for MavLink or MQTT for instance.

2.1 Installation

The Robot Diagnostic Module is distributed as a debian package. A package is available for each of the following configuration

Ubuntu Distribution	ROS1 version	ROS2 version	Architecture
22.04	N/A	Humble/Iron	amd64/arm64v8
20.04	Noetic	Galactic/Foxy	amd64/arm64v8
18.04	Melodic	N/A	amd64/arm64v8

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2.1.1 Quick Installation

Here is the one liner that will download a public script that will, based on your system download the right package.

```
sudo bash -c "$(wget -qO -
https://raw.githubusercontent.com/3LawsRobotics/3laws/master/rdm/install.sh)"
```

2.1.2 Non Interactive Installation

If you want to install a specific package in a non interactive mode, here is the process:

- Download the script:

```
wget https://raw.githubusercontent.com/3LawsRobotics/3laws/master/rdm/install.sh
```

- Make it executable

```
chmod +x install.sh
```

- Run it with your arguments

```
sudo ./install.sh [-hyf] [-r <ROS_DISTRO>] [-a <ARCH>] [-v <UBUNTU_VERSION>]
```

if `-yf -r <ROS_DISTRO> -a <ARCH> -v <UBUNTU_VERSION>` specified, the script is non interactive

2.2 Launch the RDM

To launch the RDM, you only have to source your ros distribution:

```
source /opt/ros/<DISTRO>/setup.sh
```

and to run the command:

```
ros2 launch lll_rdm rdm.launch.py
```

You can add the following command line options:

- use_sim_time: <Boolean, default: false> Specify if the RDM use the ros time or the machine time
- config_filename: <String> Path to the config file if not in the */opt/3lawsRoboticsInc/robot_diagnostic_module/config* folder (**needs to start with /**) or just the config file name inside the default folder.
- robot_id: <String> name of your robot
- log_level: <trace | debug | info | error, default: info> Log level of the RDM
- dry_run: <Boolean> Start the RDM, check the config file and turn off without error if everything has been launched correctly

2.3 Configuration

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The configuration is written inside a YAML. file. This file can be stored anywhere but the default folder is `/opt/3lawsRoboticsInc/robot_diagnostic_module/config`

Other path can be specified in the command line launch command

The Robot Diagnostic Module is shipped with a configuration template that can be found here
`/opt/3lawsRoboticsInc/robot_diagnostic_module/config/template_config.yaml`

In order to have the more user friendly configuration experience, a lot of the fields are optional. The configuration can be build alongside your robot so that it does not represent a burden for the developer.

2.3.1 General configuration rules

Bounds

All bounds parameters are similar:

- norm_type: Norm to use for norm bound, possible values (none, L_1, L_2, L_infinity)
- norm_upper_bound: Upper norm bound, floating point
- norm_lower_bound: Lower norm bound, floating point
- upper_bounds: Upper bounds on the value of the components of the tracking_error, sequence of floating point
- lower_bounds: Lower bounds on the value of the components of the tracking_error, sequence of floating point
- rates_upper_bounds: Upper bounds on the value of the components of the tracking_error, sequence of floating point
- rates_lower_bounds: Lower bounds on the value of the components of the tracking_error, sequence of floating point

Set both (rate_)upper_bounds[i] and (rates_)lower_bounds[i] to .inf to disable bound on component i.

Leave empty to deactivate all (rate) bounds checking.

Both (rate_)upper_bounds and (rates_)lower_bounds must have same size

Duration parameters

Parameters like signal_min_rate represent durations. They can be specified by an integer number corresponding to a number of nanoseconds:

signal_min_rate: 1 -> 1 nanoseconds

signal_min_rate: 1000000000 -> 1 second

They can also be specified by an integer of floating point followed by the time scale:

signal_min_rate: 1ns -> 1 nanoseconds

signal_min_rate: 25us -> 25 microseconds

signal_min_rate: 0.5ms -> 500 milliseconds

signal_min_rate: 1s -> 1 second

signal_min_rate: 20hz -> 50 milliseconds

signal_min_rate: 1min -> 1 minute

signal_min_rate: 1h -> 1 hour

signal_min_rate: 1d -> 1 day

signal_min_rate: 1y -> 1 year

2.3.2 Scheme

Credentials

Credentials are used to connect to the databases to store the data. Here is the structure and the field detail.

```
credentials:
  company_id: Company name
  robot_id: Robot Name
  influx_credentials:
    database_token: INFLUX token provided by 3lawsRobotics
  clickhouse_credentials:
    username: Company name
    password: CLICKHOUSE password provided by 3lawsRobotics
```

Diagnostic Configuration

This part of the configuration is here to define diagnostic specific options.

Legend:

- <Distance array> An array of distance in meter
- <Percentage array> An array of percentages (between 0 and 100)
- <Time array> An array of time. Allowed unit: s, ms, min
- <Ratio Array> An array of ratio (between 0 and 1). Used when values are normalized
- <Severity> An enum: ok | minor | severe | critical | ~

If a value is set to [] or ~ the RDM will not flag event for this specific case. In the case of an array option, the first value is the threshold to detect a critical event, the second for a severe event and so on. So for example, a Distance Array like this [0.001, 0.005, 0.01] will flag a minor event if the value goes under 0.01, then reevaluates it to 0.005 if the value is inferior to 0.005 and finally raise the severity to critical if the value goes under 0.001

```
diagnostic_config:
  timeout_factor: <Int> Specify how much time the max signal rate has to be wait
                        to declare a topic timed out. This trigger an event.
  max_signals_delay: <Time <int> ms/s/min> Specify the maximum delay between
                                           the send time and the received time
  incident_detection: This structure specify the severity and the threshold for
the event metrics
    min_event_time: <Time <int> ms/s/min> Minimal time to flag an event
    min_displayed_severity: <Enum: ok|minor|severe|critical> Minimal criticality
to flag an event
    collision:
      limit_dists: <Distance array>
    computer:
      limit_cpu_loads: <Percentage array>
      limit_disk_usages: <Percentage array>
      limit_ram_usages: <Percentage array>
    clock:
      limit_utc_deviations: <Time array>
      limit_rtc_deviations: <Time array>
    node_health:
      timeout_severity: <Severity>
      not_ok_severity: <Severity>
      limit_delay_bounds: <Ratio Array>
      limit_rate_bounds: <Ratio Array>
    obstruction:
      obstruction_severity: <Severity>
    signal_health:
      nan_severity: <Severity>
      zero_severity: <Severity>
      subnormal_severity: <Severity>
      inf_severity: <Severity>
      bad_timestamp_severity: <Severity>
      limit_bounds: <Ratio Array>
    dynamic:
      limit_xdot_difference: <Ratio Array>
      input_domain_severity: <Severity>
      state_domain_severity: <Severity>
    tracking:
      limit_tracking_difference: <Ratio Array>
  upload: This configuration allow the user to reduce the bandwidth
          by not uploading high frequency metrics
    high_frequency:
```

```
      state: <Boolean>
      input: <Boolean>
      path_tracking_error: <Boolean>
      control_tracking_error: <Boolean>
```

Interface Configuration

The interface structure is here to get precision on the interface. Right know only ROS interfaces are available, the interface option are here to specify Quality of Service and topic types

```
interface:
  retimestamp: never/always/if_zero specify if the timestamp has to be set

  ros_topics_info:
    /example_topic!:
      type: sensor_msgs/LaserScan
    /example_topic2:
      type: lll_msgs/Float64VectorStamped
    /example_topic3:
      type: nav_msgs/Odometry
```

Supported topics type are all ROS default types and 3lawsRobotics custom types (ns: lll_msg, lll_rdm_msg)

Robot Description

This part is the most important one. The more accurate the information provided here, the more insightful the RDM will be.

Each sub structure is optional but once a substructure is declared, all its fields must be filled (unless Optional is specified)

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Robot model

```
robot_type: <String> See available robot type section for more details

kinematic_model: # Kinematic model of the robot
  base_frame_id: <String> # Name of the base frame of the robot
  geometry: # Geometry of the robot
    shape_type: <Enum point|box|sphere|ellipsoid|capsule|cone|cylinder|mesh> #
Type of the geometry shape (should be one of [point, box, sphere, ellipsoid,
capsule, cone, cylinder, mesh])
    shape_params: # Parameters for the chosen shape_type
      x_size: <Float> #in Meter
      y_size: <Float> #in Meter
      z_size: <Float> #in Meter
    pose: # [optional] Pose of shape w.r.t base frame
      translation: <Float array[3]> # [optional] Translation w.r.t base frame
[x,y,z], default [0,0,0]
      rotation: <Float array[4]> # [optional] Rotation quaternion w.r.t base frame
[w,x,y,z], default [1,0,0,0]

dynamical_model: # [optional] Behavior model of the robot
  model_type: <String> # Type of dynamical model for selected robot_type
  state_domain: # Region of the state space in which the dynamical model has been
validated
    simple: <Boolean> # [optional], If true, assumes M is an identity matrix
```

```
(default: false)
    n_cstr: 3 # [optional if simple==true] Number of rows in M
    M: # [optional if simple==true] Either a single sequence representing
the diagonal of M, or a sequence of the rows of M
      - [1., 0., 0., 0., 0.]
      - [0., 1., 0., 0., 0.]
      - [0., 0., 1., 0., 0.]
    # Defined as a a convex polytope S={x in Rn | lb <= M.x <= ub}
    ub:
      <Float array[n_cstr]> # If simple==true, must have same size as state of
model_type,
      # otherwise must have size n_cstr
    lb:
      <Float array[n_cstr]> # If simple==true, must have same size as state of
model_type,
      # otherwise must have size n_cstr

    input_domain: # Region of the input space accessible for control purposes and
in which the dynamical model has been validated
      # Same representation as 'state_domain'
      ub:
        <Float array[model_input_size]> # If simple==true, must have same size as
input of model_type,
        # otherwise must have size n_cstr
      lb:
        <Float array[model_input_size]> # If simple==true, must have same size as
input of model_type,
        # otherwise must have size n_cstr
      simple: <Boolean>

  process_noise_covariance:
    <Float array[model_state_size]> # Either a single sequence representing
the diagonal of the process noise covariance matrix,
    # or a sequence of its rows. Must have same size as state of model_type
  model_param: # [variant] Parameters for the chosen model_type, as example here
for the bicycle
    wheel_dx: 1.
    origin_dx: 1.
    tau_vel: 1.
    tau_steer: 1.
```

Available shape types

The kinematic model of the robot defines its geometry. The geometry is described by a base frame, a shape, and the pose of that shape in the base frame. We currently support the following shapes:

- **Point**:

```
shape_params: ~
```

- **Box**:

```
shape_params:
  x_size: <Float>  Total length of the box along x axis
  y_size: <Float>  Total length of the box along y axis
  z_size: <Float>  Total length of the box along z axis_mask
```

- **Sphere**:

```
shape_params:
  radius: <Float>  Radius of the sphere
```

- **Ellipsoid**:

```
shape_params:
  radius_x: <Float>  Semi x-axis length
  radius_y: <Float>  Semi y-axis length
  radius_z: <Float>  Semi z-axis length
```

- **Capsule**:

```
shape_params:
  radius: <Float>  Radius of the capsule
  length: <Float>  Length of the capsule
```

- **Cone**:

```
shape_params:
  radius: <Float>  Radius of the cone
  length: <Float>  Length of the cone
```

- **Cylinder**:

```
shape_params:
  radius: <Float>  Radius of the cylinder
  length: <Float>  Length of the cylinder
```

- **Mesh**:

```
shape_params:
  mesh_file: /opt/mesh.stl  Path to mesh file
  mesh_type: stl  Type of mesh file, available options: [stl]
  mesh_units: mm  [optional] Units of the mesh file, available options: [mm, cm,
dm, m, dam, hm, km, mi, nm, yd, ft, in], default: m
```

Available robot type

Each dynamical model type has its own set of states, inputs, and parameters:

- mobile_robot:
  - differential_drive: Dynamical model for a rigid body over SE2 with first order tracking response of longitudinal and rotational body velocities
    states: [x,y,yaw,vx_body_actual,wz_body_actual]
    inputs: [vx_body_command,wz_body_command]
    parameters:
    - tau_vel: time constant of the 1st order tracking response in linear velocity (1/s) (must be strictly positive)
    - tau_yaw_vel: time constant of the 1st order tracking response in angular velocity (1/s) (must be strictly positive)
  - bicycle: Dynamical model for a 2 wheels or 4 wheel but with coupled front wheel steering vehicle over SE2, with first order tracking response of steering angle and origin velocity magnitude.
    states: [x,y,yaw,||v_body||_actual,steering_angle_actual]
    inputs: [||v_body||_command,steering_angle_command]
    parameters:
    - wheel_dx: Distance between front and back wheels (m) (must be strictly positive)
    - origin_dx: Position of vehicle's origin w.r.t back wheels (m) (must be positive)

- tau_vel: time constant of the 1st order tracking response in linear velocity (1/s) (must be strictly positive)
- tau_steer: time constant of the 1st order tracking response in angular velocity (1/s) (must be strictly positive)

Auto Generated Nodes

The RDM will automatically create node based on the configuration. So for each substructure of robot_description, a node will be created. For example a node sensor is created and will group all topics from each sensor declared in the configuration.

If you want to add extra topic to a node, you can declare them via the **extra_topics** structure. This structure is a list of topics.

A topic has to be declared like this:

```
- interface_id:
    /test_node_topic_1 # Name of the ros topic. Associated
'interface.message_type_map.<interface_id>' must be specified.
    # Supported types: [builtin_interfaces/*, geometry_msgs/*, lll_msgs/*,
nav_msgs/*, rcl_interfaces/*, rosgraph_msgs/*, sensor_msgs/*, std_msgs/*,
trajectory_msgs/*, visualization_msgs/*]
  topic_id: test_node_topic_1 # Display name for this topic, must be UNIQUE among
all topics of each node
  signal_min_rate: 1s # Maximum allowed duration without receiving data
```

If a topic is declared like this, inside the interface section, **the message_type has to be declared**.

Another information can be added to a node: its process_name. When a process name is specified, the RDM will find the process and monitor his CPU and network usage.

Robot Autonomy stack

Mission Manager

```
mission_manager:
  extra_topics: ~ # [optional]
  process_name: ~ # [optional]
  finite_states: # Finite states of the robot
    - interface_id:
        /status # Name of the ros topic.
        # Supported types: [(default) std_msgs/String, std_msgs/FloatXX,
std_msgs/Bool, std_msgs/Char, std_msgs/Byte, std_msgs/IntXX, std_msgs/UIntXX]
      sender_id: state_machine # Display name for sender of this state
      state_id: status # Identifier for this state, "<sender_id>.<state_id>" must
form a UNIQUE identifier among all signals
      signal_min_rate: 1s
    - interface_id: /search_mode
      sender_id: state_machine
      state_id: search_mode
      signal_min_rate: 1s
```

Path Planning

```
path_planning:
  extra_topics: ~ # [optional]
  process_name: ~ # [optional]
  paths:
    - interface_id:
        /desired_path # Name of the ros topic.
        # Supported types: [(default) lll_msgs/Trajectory, nav_msgs/Path,
trajectory_msgs/JointTrajectory]
      path_id: main_path # Display name for this path, must be UNIQUE among all
paths
      trajectory_state_size: 7 # Size of the trajectory state vector
```

```
      signal_min_rate: 1min # Maximum time without receiving data before signal is
considered timed out
      # state_mask:
      #   [0, 1, 2, 3, 4, 5, 6] # [optional] If the path only corresponds to
a subset of the state_estimation vector,
      #   # use this mask to extract the relevant data : trajectory_state[i] =
state_estimation[state_mask[i]].
      #   # Must be of size 'trajectory_state_size', and not contain indices
greater than state_estimation.state_size.
      #   # If not specified or null, will be [0, ..., trajectory_state_size-1]
      tracking_error_bounds: ~ # [optional] Bounds on controller's tracking error
: path_state - actual_state
```

Robot Perception

Sensors:

```
sensors:
    extra_topics: ~ # [optional]
    process_name: ~ # [optional]
    batteries: [] # Coming soon!
    cameras: [] # Coming soon!
    gps: [] # Coming soon!
    imus: [] # Coming soon!
    lidars: [] # Coming soon!
    laserscans: # See next section
    loadcells: # See next section
```

Laserscan

```
laserscans: # Planar laser scanners
  - interface_id:
      /laserscan_1_topic # Name of the ros topic.
      # Supported types: [(default) sensor_msgs/LaserScan]
    sensor_id: laserscan_1 # Display name for this laserscan, must be UNIQUE among
all laserscans
    signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
    specs:
      n_rays: 1000 # Expected number of rays in the laserscan
      angle_min: -3.14 # Minimum ray angle
      angle_max: 3.14 # Maximum ray angle
      range_min: 0. # Minimum ray range
      range_max: 1000. # Maximum ray range
      noise_one_sigma: 0.025 # Expected standard_error of the sensor (given by
the manufacturer, often like: precision = +-2sigma)
    transform: # Specification of frame w.r.t which the measurement is expressed
      parent_frame_id: robot # Id of parent frame
      pose: # [optional] Pose w.r.t parent frame
        translation: [0., 0., 0.] # [optional] Translation w.r.t parent frame
[x,y,z], default [0,0,0]
        rotation: [1., 0., 0., 0.] # [optional] Rotation quaternion w.r.t parent
frame [w,x,y,z], default [1,0,0,0]
```

Loadcells

```
loadcells: # Force and torque measurement sensor, 6 axis by default
  - interface_id:
      /end_effector_wrench # Name of the ros topic.
      # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
vectorizable type (see bottom of this file)]
    sensor_id: end_effector_loadcell # Display name for this loadcell, must be
UNIQUE among all loadcells
    signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
```

```
    transform: # Specification of frame w.r.t which the measurement is expressed
      parent_frame_id: robot # Id of parent frame
      pose: # [optional] Pose w.r.t parent frame
        translation: [0., 0., 0.] # [optional] Translation w.r.t parent frame
[x,y,z], default [0,0,0]
        rotation: [1., 0., 0., 0.] # [optional] Rotation quaternion w.r.t parent
frame [w,x,y,z], default [1,0,0,0]
    # axis_mask: # [optional] Define which of the 6 force/torque axes in SE3
the loadcell signals correspond to: [Fx, Fy, Fx, Mx, My, Mz].
    #   # If not specified or null, assumes all 6 axes.
    #   # Cannot be empty or longer than 6. Index must be between 0 and 5
included.
    #   [0, 5] # Corresponds to a 2 axis loadcell [Fx,Mz]
    noise_one_sigma: [1., 1., 1., 1., 1., 1.] # Noise characteristics of loadcell
axes. Must have same size as axis_mask
    bounds: ~ # [optional]
```

Perception

```
perception:
  obstacles: # [optional] List of obstacles
    interface_id: /obstacles # Name of the ros topic. # Supported types:
[(default) lll_msgs/ObjectArray]
    signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
    meshes: # List of meshes to be loaded
      []
      # - id: sphere # Mesh identifier, must be UNIQUE among all meshes
      #   data: # Mesh data
      #     mesh_file: sphere.stl # Path to mesh file
      #     mesh_type: stl # Type of mesh file
      #     mesh_units: mm # Unit of mesh file
```

Localization

```
localization:
  extra_topics: ~ # [optional]
  process_name: ~ # [optional]
  state_estimation: # [optional] See next section
  odometry: # [optional] See next section
```

State Estimation

```
state_estimation:
  interface_id:
    /state # Name of the ros topic.
    # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
vectorizable type (see bottom of this file)]
  signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
  state_size: 5 # Size of the state vector
  # mask:
  #   [0, 1, 2, 3, 5] # [optional] If only a subset of the vectorized message
actually constitute the state vector
  #   # use this mask to extract the relevant data : state[i] =
msg_vectorized[mask[i]].
  #   # Must be of size 'state_size', and not contain indices greater than
the size of vectorized message.
  #   # If not specified or null, will be [0, ..., state_size-1]
  bounds: ~ # [optional]
```

Odometry

```
odometry:
  - interface_id:
```

```
      /odom_node_0_topic # Name of the ros topic.
      # Supported types: [(default) nav_msgs/Odometry]
    odom_id: odom_node_0 # Display name for this odometry source, must be UNIQUE
among all odometry
    signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
    se2_only: true # [optional] Consider only SE2 projection of pose and twist
(default: false)
    position_bounds: # [optional] Position part of the odometry. Components are
[x,y,z] or [x,y] if se2_only==true
      norm_type: none
      norm_upper_bound: 1.
      norm_lower_bound: 0.
      upper_bounds: [1., 1.]
      lower_bounds: [-1., -1.]
      rates_upper_bounds: []
      rates_lower_bounds: []
    orientation_bounds: ~ # [optional] Same fields as position. Components are
[roll,pitch,yaw] or [yaw] if se2_only==true
    velocity_linear_bounds: ~ # [optional] Same fields as position. Components are
[vx,vy,vz] or [vz,vy] if se2_only==true
    velocity_angular_bounds: ~ # [optional] Same fields as position. Components
are [wx,wy,wz] or [wz] if se2_only==true
```

Robot control

```
control:
  extra_topics: ~ # [optional]
  process_name: ~ # [optional]
  setpoint_tacking_controllers: # See next section
  actuators: # See next section
  supervisors: # See next section
```

SetPoint Tacking Controller

```
setpoint_tacking_controllers: # PID like controllers
  - controller_id: velocity_controller # Display name for this controller, must be
UNIQUE among all controllers
    state_size: 1 # Size of controller setpoint
    input_size: 1 # Size of control input computed by controller
    desired_state:
      interface_id: /controller_cmd_topic # Name of the desired state ros topic.
      # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
vectorizable type (see bottom of this file)]
      mask: [0] # [optional] If only a subset of desired_state_topic_id vector is
actually used by controller,
      # use this mask to extract the relevant data : desired_state_used[i] =
desired_state_received[desired_state_mask[i]]
      signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
      bounds: ~ # [optional] Bounds on desired state

    actual_state:
      interface_id: /controller_state_topic # Name of the actual state ros topic.
      # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
vectorizable type (see bottom of this file)]
      mask: ~ # [optional] Same as desired_state_mask
      signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
      bounds: ~ # [optional] Bounds on actual state

    control_input:
      interface_id: /controller_input_topic # Name of the control input ros topic.
      # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
```

```
vectorizable type (see bottom of this file)]
      mask: ~ # [optional] Same as desired_state_mask
      signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
      bounds: # [optional] Bounds on controller's control input
        norm_type: none
        norm_upper_bound: 1.
        norm_lower_bound: 0.
        upper_bounds: []
        lower_bounds: []
        rates_upper_bounds: [1.]
        rates_lower_bounds: [-1.]

    tracking_error_bounds: # [optional] Bounds on controller's tracking error :
desired_state - actual_state
      norm_type: none
      norm_upper_bound: 1.
      norm_lower_bound: 0.
      upper_bounds: [1.]
      lower_bounds: [-1.]
      rates_upper_bounds: []
      rates_lower_bounds: []
```

Actuators

```
actuators: # Robot actuation
  combined: # Combined actuation vector
    interface_id:
      /control_input # Name of the ros topic publishing the complete robot
actuation vector.
      # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
vectorizable type (see bottom of this file)]
    input_size: 3 # Size of the combined input vector
    signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
    bounds: ~ # [optional]
    # mask:
    #   [0,1,3] # [optional] If only a subset of the vectorized message actually
constitute the combined input vector
    #   # use this mask to extract the relevant data : input[i] =
msg_vectorized[mask[i]].
    #   # Must be of size 'input_size', and not contain indices greater than
the size of the vectorized message.
    #   # If not specified or null, will be [0, ..., input_size-1]
```

Supervisors

```
supervisors: # 3Laws AI Supervisors
  - interface_id: /main_supervisor_topic # Name of the supervisor data ros topic.
    supervisor_id: main_supervisor # Display name for this supervisor, must be
UNIQUE among all supervisor
    signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
```

Extras

```
extras:
  passthrough_metrics: # See next section
  clocks: # See next section
  signals: # See next section
  nodes: # See next section
```

Passthrough Metrics

```
passthrough_metrics: # Generic passthrough for scalar metric signals
```

```
  - interface_id:
      /metric_1_topic # Name of the ros topic.
      # Supported types: [(default) std_msgs/Float64, std_msgs/Float32,
std_msgs/Bool, std_msgs/Char, std_msgs/Byte, std_msgs/IntXX, std_msgs/UIntXX]
    metric_id: metric_1 # Display name for this metric, must be UNIQUE among all
passthrough metrics
    metric_group_id:
      position # [optional] Group this signal belongs to.
      # Metrics of the same group are plotted on the same graph in 3laws.app
```

Clocks

```
clocks:
  - interface_id:
      /custom_clock # Name of the ros topic.
      # Supported types: [(default) rosgraph_msgs/Clock]
    clock_id: my_clock # Display name for this clock, must be UNIQUE among all
clocks
    signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
```

Signals

```
signals: # Generic floating point multidimensional signal values sanity and bounds
checking
  - interface_id:
      /test_signal_topic # Name of the ros topic.
      # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
vectorizable type (see bottom of this file)]
    sender_id: test_signal_node # Display name of sender node
    signal_id: test_signal # Display name of this signal,
"<sender_id>.<signal_id>" must form a UNIQUE identifier among all signals
    signal_size: 1 # Size of this signal
    signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
    # mask:
    #   [2] # [optional] If only a subset of vectorized message actually
constitute the signal vector
    #   # use this mask to extract the relevant data : signal[i] =
msg_vectorized[mask[i]].
    #   # Must be of size 'signal_size', and not contain indices greater than
the size of the vectorized message.
    #   # If not specified or null, will be [0, ..., signal_size-1]
    bounds: ~ # [optional]
```

Nodes

```
nodes: # Generic node health checking metric
  - node_id: test_node # Display name of node, must be UNIQUE among all nodes
    # text_log_interface_id:
    #   /test_node_log # [optional] Name of the ros topic publishing log info for
that node.
    #   # Supported types: [(default) rcl_interfaces/Log]
    # process_name: # [optional]
    #   test_node_exec.
    topics: # List of topics published by the node (only available in ros2 humble
and up)
      - interface_id:
          /test_node_topic_1 # Name of the ros topic. Associated
'interface.message_type_map.<interface_id>' must be specified.
          # Supported types: [builtin_interfaces/*, geometry_msgs/*, lll_msgs/*,
nav_msgs/*, rcl_interfaces/*, rosgraph_msgs/*, sensor_msgs/*, std_msgs/*,
trajectory_msgs/*, visualization_msgs/*]
        topic_id: test_node_topic_1 # Display name for this topic, must be UNIQUE
```

```
among all topics of each node
        signal_min_rate: 1s # Maximum allowed duration without receiving data
  - node_id: rosout
    text_log_interface_id: /rosout # If equal to '/rosout', uses 'name' field of
incoming rcl_interfaces/Log message as node_id for text_log message
    topics: []
```

Template configuration

You can copy and paste this configuration and change the value to suit your robot characteristic

```
# Credentials
credentials:
  company_id: COMPANY_ID # Your company identifier
  robot_id: "" # Your robot identifier
  influx_credentials:
    database_token: TOKEN # Your company database token
  clickhouse_credentials:
    username: CLICKHOUSE_USR
    password: CLICKHOUSE_PWD
    database: CLICKHOUSE_DATABASE

# Robot description - Definition of expected behavior of your system
robot_description:
  robot_type: mobile_robot

  kinematic_model: # Kinematic model of the robot
    base_frame_id: robot # Name of the base frame of the robot
    geometry: # Geometry of the robot
      shape_type: box # Type of the geometry shape (should be one of [point, box,
sphere, ellipsoid, capsule, cone, cylinder, mesh])
      shape_params: # Parameters for the chosen shape_type (See bottom of this
file for more info)
        x_size: 0.5
        y_size: 0.4
        z_size: 0.3
      pose: # [optional] Pose of shape w.r.t base frame
        translation: [0., 0., 0.] # [optional] Translation w.r.t base frame
[x,y,z], default [0,0,0]
        rotation: [1., 0., 0., 0.] # [optional] Rotation quaternion w.r.t base
frame [w,x,y,z], default [1,0,0,0]

  dynamical_model: # [optional] Behavior model of the robot
    model_type: bicycle # Type of dynamical model for selected robot_type
    state_domain: # Region of the state space in which the dynamical model has
been validated
      # Defined as a a convex polytope S={x in Rn | lb <= M.x <= ub}
      ub:
        [1., 2., 3.] # If simple==true, must have same size as state of
model_type,
        # otherwise must have size n_cstr
      lb:
        [-1., -2., -3.] # If simple==true, must have same size as state of
model_type,
        # otherwise must have size n_cstr
      simple: false # [optional], If true, assumes M is an identity matrix
(default: false)
      n_cstr: 3 # [optional if simple==true] Number of rows in M
      M: # [optional if simple==true] Either a single sequence representing
the diagonal of M, or a sequence of the rows of M
        - [1., 0., 0., 0., 0.]
        - [0., 1., 0., 0., 0.]
        - [0., 0., 1., 0., 0.]
    input_domain: # Region of the input space accessible for control purposes and
```

```
in which the dynamical model has been validated
      # Same representation as 'state_domain'
      ub:
        [1., 2.] # If simple==true, must have same size as input of model_type,
        # otherwise must have size n_cstr
      lb:
        [-1., -2.] # If simple==true, must have same size as input of model_type,
        # otherwise must have size n_cstr
      simple: true
    process_noise_covariance:
      [1., 1., 1., 1., 1.] # Either a single sequence representing the diagonal of
the process noise covariance matrix,
      # or a sequence of its rows. Must have same size as state of model_type
    model_param: # [variant] Parameters for the chosen model_type (See bottom of
this file for more info)
      wheel_dx: 1.
      origin_dx: 1.
      tau_vel: 1.
      tau_steer: 1.

  mission_manager: # [optional]
    extra_topics: ~ # [optional] (See bottom of this file for more info)
    process_name: ~ # [optional] (See bottom of this file for more info)
    finite_states: # Finite states of the robot
      - interface_id:
          /status # Name of the ros topic.
          # Supported types: [(default) std_msgs/String, std_msgs/FloatXX,
std_msgs/Bool, std_msgs/Char, std_msgs/Byte, std_msgs/IntXX, std_msgs/UIntXX]
        sender_id: state_machine # Display name for sender of this state
        state_id: status # Identifier for this state, "<sender_id>.<state_id>"
must form a UNIQUE identifier among all signals
        signal_min_rate: 1s
      - interface_id: /search_mode
        sender_id: state_machine
        state_id: search_mode
        signal_min_rate: 1s

  sensors: # [optional]
    extra_topics: ~ # [optional] (See bottom of this file for more info)
    process_name: ~ # [optional] (See bottom of this file for more info)
    batteries: [] # Coming soon!
    cameras: [] # Coming soon!
    gps: [] # Coming soon!
    imus: [] # Coming soon!
    laserscans: # Planar laser scanners
      - interface_id:
          /laserscan_1_topic # Name of the ros topic.
          # Supported types: [(default) sensor_msgs/LaserScan]
        sensor_id: laserscan_1 # Display name for this laserscan, must be UNIQUE
among all laserscans
        signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
        specs:
          n_rays: 1000 # Expected number of rays in the laserscan
          angle_min: -3.14 # Minimum ray angle
          angle_max: 3.14 # Maximum ray angle
          range_min: 0. # Minimum ray range
          range_max: 1000. # Maximum ray range
          noise_one_sigma: 0.025 # Expected standard_error of the sensor (given by
the manufacturer, often like: precision = +-2sigma)
        transform: # Specification of frame w.r.t which the measurement is
expressed
          parent_frame_id: robot # Id of parent frame
```

```
          pose: # [optional] Pose w.r.t parent frame
            translation: [0., 0., 0.] # [optional] Translation w.r.t parent frame
[x,y,z], default [0,0,0]
            rotation: [1., 0., 0., 0.] # [optional] Rotation quaternion w.r.t
parent frame [w,x,y,z], default [1,0,0,0]
    lidars: [] # Coming soon!
    loadcells: # Force and torque measurement sensor, 6 axis by default
      - interface_id:
          /end_effector_wrench # Name of the ros topic.
          # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
vectorizable type (see bottom of this file)]
        sensor_id: end_effector_loadcell # Display name for this loadcell, must be
UNIQUE among all loadcells
        signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
        transform: # Specification of frame w.r.t which the measurement is
expressed
          parent_frame_id: robot # Id of parent frame
          pose: # [optional] Pose w.r.t parent frame
            translation: [0., 0., 0.] # [optional] Translation w.r.t parent frame
[x,y,z], default [0,0,0]
            rotation: [1., 0., 0., 0.] # [optional] Rotation quaternion w.r.t
parent frame [w,x,y,z], default [1,0,0,0]
        # axis_mask: # [optional] Define which of the 6 force/torque axes in SE3
the loadcell signals correspond to: [Fx, Fy, Fx, Mx, My, Mz].
        #   # If not specified or null, assumes all 6 axes.
        #   # Cannot be empty or longer than 6. Index must be between 0 and 5
included.
        #   [0, 5] # Corresponds to a 2 axis loadcell [Fx,Mz]
        noise_one_sigma: [1., 1., 1., 1., 1., 1.] # Noise characteristics of
loadcell axes. Must have same size as axis_mask
        bounds: ~ # [optional] (See bottom of this file for more info)

  perception: # [optional]
    obstacles: # [optional] List of obstacles
      interface_id: /obstacles # Name of the ros topic. # Supported types:
[(default) lll_msgs/ObjectArray]
      signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
      meshes: # List of meshes to be loaded
        []
        # - id: sphere # Mesh identifier, must be UNIQUE among all meshes
        #   data: # Mesh data
        #     mesh_file: sphere.stl # Path to mesh file
        #     mesh_type: stl # Type of mesh file
        #     mesh_units: mm # Unit of mesh file

  localization: # [optional]
    extra_topics: ~ # [optional] (See bottom of this file for more info)
    process_name: ~ # [optional] (See bottom of this file for more info)
    state_estimation: # [optional]
      interface_id:
        /state # Name of the ros topic.
        # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
vectorizable type (see bottom of this file)]
      signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
      state_size: 5 # Size of the state vector
      # mask:
      #   [0, 1, 2, 3, 5] # [optional] If only a subset of the vectorized message
actually constitute the state vector
      #   # use this mask to extract the relevant data : state[i] =
msg_vectorized[mask[i]].
      #   # Must be of size 'state_size', and not contain indices greater than
```

```
the size of vectorized message.
      #   # If not specified or null, will be [0, ..., state_size-1]
      bounds: ~ # [optional] (See bottom of this file for more info)

    odometry:
      - interface_id:
          /odom_node_0_topic # Name of the ros topic.
          # Supported types: [(default) nav_msgs/Odometry]
        odom_id: odom_node_0 # Display name for this odometry source, must be
UNIQUE among all odometry
        signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
        se2_only: true # [optional] Consider only SE2 projection of pose and twist
(default: false)
        position_bounds: # [optional] Position part of the odometry. Components
are [x,y,z] or [x,y] if se2_only==true (See bottom of this file for more info)
          norm_type: none
          norm_upper_bound: 1.
          norm_lower_bound: 0.
          upper_bounds: [1., 1.]
          lower_bounds: [-1., -1.]
          rates_upper_bounds: []
          rates_lower_bounds: []
        orientation_bounds: ~ # [optional] Same fields as position. Components are
[roll,pitch,yaw] or [yaw] if se2_only==true (See bottom of this file for more
info)
        velocity_linear_bounds: ~ # [optional] Same fields as position. Components
are [vx,vy,vz] or [vz,vy] if se2_only==true (See bottom of this file for more
info)
        velocity_angular_bounds: ~ # [optional] Same fields as position.
Components are [wx,wy,wz] or [wz] if se2_only==true (See bottom of this file for
more info)

  path_planning: # [optional]
    extra_topics: ~ # [optional] (See bottom of this file for more info)
    process_name: ~ # [optional] (See bottom of this file for more info)
    paths:
      - interface_id:
          /desired_path # Name of the ros topic.
          # Supported types: [(default) lll_msgs/Trajectory, nav_msgs/Path,
trajectory_msgs/JointTrajectory]
        path_id: main_path # Display name for this path, must be UNIQUE among all
paths
        trajectory_state_size: 7 # Size of the trajectory state vector
        signal_min_rate: 1min # Maximum time without receiving data before signal
is considered timed out
        # state_mask:
        #   [0, 1, 2, 3, 4, 5, 6] # [optional] If the path only corresponds to
a subset of the state_estimation vector,
        #   # use this mask to extract the relevant data : trajectory_state[i] =
state_estimation[state_mask[i]].
        #   # Must be of size 'trajectory_state_size', and not contain indices
greater than state_estimation.state_size.
        #   # If not specified or null, will be [0, ..., trajectory_state_size-1]
        tracking_error_bounds: ~ # [optional] Bounds on controller's tracking
error : path_state - actual_state (See bottom of this file for more info)

  control: # [optional]
    extra_topics: ~ # [optional] (See bottom of this file for more info)
    process_name: ~ # [optional] (See bottom of this file for more info)
    setpoint_tacking_controllers: # PID like controllers
      - controller_id: velocity_controller # Display name for this controller,
must be UNIQUE among all controllers
```

```
        state_size: 1 # Size of controller setpoint
        input_size: 1 # Size of control input computed by controller
        desired_state:
          interface_id: /controller_cmd_topic # Name of the desired state ros
topic.
          # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
vectorizable type (see bottom of this file)]
          mask: [0] # [optional] If only a subset of desired_state_topic_id vector
is actually used by controller,
          # use this mask to extract the relevant data : desired_state_used[i] =
desired_state_received[desired_state_mask[i]]
          signal_min_rate: 1s # Maximum time without receiving data before signal
is considered timed out
          bounds: ~ # [optional] Bounds on desired state (See bottom of this file
for more info)

        actual_state:
          interface_id: /controller_state_topic # Name of the actual state ros
topic.
          # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
vectorizable type (see bottom of this file)]
          mask: ~ # [optional] Same as desired_state_mask
          signal_min_rate: 1s # Maximum time without receiving data before signal
is considered timed out
          bounds: ~ # [optional] Bounds on actual state (See bottom of this file
for more info)

        control_input:
          interface_id: /controller_input_topic # Name of the control input ros
topic.
          # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
vectorizable type (see bottom of this file)]
          mask: ~ # [optional] Same as desired_state_mask
          signal_min_rate: 1s # Maximum time without receiving data before signal
is considered timed out
          bounds: # [optional] Bounds on controller's control input (See bottom of
this file for more info)
            norm_type: none
            norm_upper_bound: 1.
            norm_lower_bound: 0.
            upper_bounds: []
            lower_bounds: []
            rates_upper_bounds: [1.]
            rates_lower_bounds: [-1.]
        tracking_error_bounds: # [optional] Bounds on controller's tracking error
: desired_state - actual_state (See bottom of this file for more info)
          norm_type: none
          norm_upper_bound: 1.
          norm_lower_bound: 0.
          upper_bounds: [1.]
          lower_bounds: [-1.]
          rates_upper_bounds: []
          rates_lower_bounds: []

    actuators: # Robot actuation
      combined: # Combined actuation vector
        interface_id:
          /control_input # Name of the ros topic publishing the complete robot
actuation vector.
          # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
vectorizable type (see bottom of this file)]
        input_size: 3 # Size of the combined input vector
        signal_min_rate: 1s # Maximum time without receiving data before signal is
```

```
considered timed out
        bounds: ~ # [optional] (See bottom of this file for more info)
        # mask:
        #   [0,1,3] # [optional] If only a subset of the vectorized message
actually constitute the combined input vector
        #   # use this mask to extract the relevant data : input[i] =
msg_vectorized[mask[i]].
        #   # Must be of size 'input_size', and not contain indices greater than
the size of the vectorized message.
        #   # If not specified or null, will be [0, ..., input_size-1]

    supervisors: # 3Laws AI Supervisors
      - interface_id: /main_supervisor_topic # Name of the supervisor data ros
topic.
        supervisor_id: main_supervisor # Display name for this supervisor, must be
UNIQUE among all supervisor
        signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out

  extras: # [optional]
    passthrough_metrics: # Generic passthrough for scalar metric signals
      - interface_id:
          /metric_1_topic # Name of the ros topic.
          # Supported types: [(default) std_msgs/Float64, std_msgs/Float32,
std_msgs/Bool, std_msgs/Char, std_msgs/Byte, std_msgs/IntXX, std_msgs/UIntXX]
        metric_id: metric_1 # Display name for this metric, must be UNIQUE among
all passthrough metrics
        metric_group_id:
          position # [optional] Group this signal belongs to.
          # Metrics of the same group are plotted on the same graph in 3laws.app

    clocks:
      - interface_id:
          /custom_clock # Name of the ros topic.
          # Supported types: [(default) rosgraph_msgs/Clock]
        clock_id: my_clock # Display name for this clock, must be UNIQUE among all
clocks
        signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out

    signals: # Generic floating point multidimensional signal values sanity and
bounds checking
      - interface_id:
          /test_signal_topic # Name of the ros topic.
          # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
vectorizable type (see bottom of this file)]
        sender_id: test_signal_node # Display name of sender node
        signal_id: test_signal # Display name of this signal,
"<sender_id>.<signal_id>" must form a UNIQUE identifier among all signals
        signal_size: 1 # Size of this signal
        signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
        # mask:
        #   [2] # [optional] If only a subset of vectorized message actually
constitute the signal vector
        #   # use this mask to extract the relevant data : signal[i] =
msg_vectorized[mask[i]].
        #   # Must be of size 'signal_size', and not contain indices greater than
the size of the vectorized message.
        #   # If not specified or null, will be [0, ..., signal_size-1]
        bounds: ~ # [optional] (See bottom of this file for more info)

    nodes: # Generic node health checking metric
```

```
      - node_id: test_node # Display name of node, must be UNIQUE among all nodes
        # text_log_interface_id:
        #   /test_node_log # [optional] Name of the ros topic publishing log info
for that node.
        #   # Supported types: [(default) rcl_interfaces/Log]
        # process_name: # [optional] (See bottom of this file for more info)
        #   test_node_exec.
        topics: # List of topics published by the node (only available in ros2
humble and up)
          - interface_id:
              /test_node_topic_1 # Name of the ros topic. Associated
'interface.message_type_map.<interface_id>' must be specified.
              # Supported types: [builtin_interfaces/*, geometry_msgs/*,
lll_msgs/*, nav_msgs/*, rcl_interfaces/*, rosgraph_msgs/*, sensor_msgs/*,
std_msgs/*, trajectory_msgs/*, visualization_msgs/*]
            topic_id: test_node_topic_1 # Display name for this topic, must be
UNIQUE among all topics of each node
            signal_min_rate: 1s # Maximum allowed duration without receiving data
      - node_id: rosout
        text_log_interface_id: /rosout # If equal to '/rosout', uses 'name' field
of incoming rcl_interfaces/Log message as node_id for text_log message
        topics: []

# Diagnostic config - Configuration for behavior of diagnostic module
diagnostic_config:
  timeout_factor: 2 # Multiplication factor on signal_min_rate to consider signal
has timeout
  max_signals_delay: 10ms # Maximum allowed delay between message being sent and
received
  incident_detection: # This configuration will define how incident are flagged
and with which severity (ok,minor,severe,critical).
    # If the field ends with severity it means that this incident is a either true
or false and will be flagged if the field is not null
    # If a field is an array it means that there is a continuous value that is
monitored and the incidents is flagged only if a threshold is outrun
    # The severity comes from the position of the threshold in the array, so index
zero is a critical event, index one is severe, and so on.
    # To not flag a event, just put a null value to the corresponding field (~) or
an empty array ([]).
    # The min displayed severity is a way to not change all the config if events
with a certain level of criticity need to be flagged,
    # but some less critical event are configured.
    min_event_time: 1s # Minimum incident duration that will be sent to
the dashboard
    min_displayed_severity: minor # Minimum incident severity level that will be
sent to the dashboard (ok,minor,severe,critical)
    collision:
      limit_dists: [] # Threshold values in meter that will trigger critical,
severe, minor severity incident. Maximum length = 4, keep empty if you don't want
to search for this event
    computer:
      limit_cpu_loads: [] # Threshold values in percentage (0-100) that will
trigger critical, severe, minor severity incident. Maximum length = 4, keep empty
if you don't want to search for this event
      limit_disk_usages: [] # Threshold values in percentage (0-100) that will
trigger critical, severe, minor severity incident. Maximum length = 4, keep empty
if you don't want to search for this event
      limit_ram_usages: [] # Threshold values in percentage (0-100) that will
trigger critical, severe, minor severity incident. Maximum length = 4, keep empty
if you don't want to search for this event
    clock:
      limit_utc_deviations: [] # Threshold values as string (100ns, 1s, 1min) that
will trigger critical, severe, minor severity incident. Maximum length = 4, keep
```

```
empty if you don't want to search for this event
      limit_rtc_deviations: [] # Threshold values as string (100ns, 1s, 1min) that
will trigger critical, severe, minor severity incident. Maximum length = 4, keep
empty if you don't want to search for this event
    node_health:
      timeout_severity: severe # Severity of an event of this type
      not_ok_severity: minor
      limit_delay_bounds: [1., 0.9, 0.5]
      limit_rate_bounds: [1., 0.9, 0.5]
    obstruction:
      obstruction_severity: ~ # Keep as null if you don't want to search for this
type of event
    signal_health:
      nan_severity: ~
      zero_severity: ~
      subnormal_severity: ~
      inf_severity: ~
      bad_timestamp_severity: severe
      limit_bounds: [0.1]
    dynamic:
      limit_xdot_difference: [0.5, 0.2]
      input_domain_severity: severe
      state_domain_severity: severe
    tracking:
      limit_tracking_difference: [6.]
  upload: # [optional] Data upload options
    high_frequency: # High frequency data upload
      state: true # System state signal
      input: true # System input signal
      path_tracking_error: false # Path tracking signal
      control_tracking_error: false # Control Tracking signal

interface:
  retimestamp: never # Re-timestamp incoming messages at time of reception.
Options: [never, if_zero, always]
  project_to_se2: true # During vectorization, project all SE3 messages to SE2,
like nav_msgs/Odometry -> [x,y,theta,vx,vy,wz]

  # Map from interface_id to info for that topic
  ros_topics_info:
    /test_signal_topic:
      type: lll_msgs/Float64VectorStamped # [optional] Format must be
package_name/MessageType, case is important!
      qos: SystemDefaultsQoS # [optional] For ROS2 only. Possible values
[SystemDefaultsQoS, ClockQoS(ros2 humble and up), SensorDataQoS, ParametersQoS,
ServicesQoS, ParameterEventsQoS, RosoutQoS(ros2 humble and up)]
    /rosout:
      qos: RosoutQoS
    /test_node_topic_1:
      type: geometry_msgs/TransformStamped
    /test_node_topic_2:
      type: geometry_msgs/PoseStamped
```

The configuration has **4** distinct parts

- Credentials:
  This part is here to specify the credentials to access the databases. These credentials are provided to you by email once your 3LawsRobotics account is created.
- Diagnostic Configuration:
  This part is the more important one. It describe your robot desired behavior, the interface data stream access for the RDM to monitor.
- Interface Configuration:

Here are specified the diagnostic specific options such has the criticality of a certain event or the threshold above which a incident is flagged.

- Robot Description:
  And this last part allow the user to specify interface specific options such has ROS Quality of Service.

- **doc** *examples/template_example*:
  You can find here a complete template configuration.

Chapter 3

The Supervisor/Copilot

3.1 Theory of Operation

The 3Laws Copilot is based on Control Barrier Functions as described in:

Backup Control Barrier Functions: Formultation and Comparative Study, Yuxiao Chen1 , Mrdjan Jankovic2 , Mario Santillo2 , and Aaron D. Ames1, arXiv:2104.11332v1 [eess.SY] 22 Apr 2021, **Applied Safety Critical Control** Thomas Gurriet, Doctor of Philosophy, CALIFORNIA INSTITUTE OF TECHNOLOGY, Pasadena, California 2020

The basic concept is to use the current state of a dynamical system (robot arm, mobile device, aircraft, marine vessel, etc.) to drive a model of that system. Inputs including locations, geometries, speeds, and accelerations of obstacles are also needed when the CoPilot is designed for collision avoidance. A prediction of when a collision will occur is used as a basis to modify the currently requested steering/speed/attitude commands when necessary. The CoPilot uses an alternative planning/control strategy to prevent the collision by slowing or diverting the device away from the collision path. The alternative control strategy is built into the CoPilot by 3Laws based on the objective for the particular deployment.

3.2 Basic Architecture

From an operational standpoint, the CoPilot (when used for collision avoidance) sits between the planning layer and the hardware control layer. "Hardware control" typically refers to a speed or attitude controller for a vehicle or a joint attitude/speed controller for an articulated robot.

The CoPilot will evaluate the likelihood of a collision and will only modify the desired input from the planner when a collision is predicted within the estimation window.